Synthetic data. Built by Russell Harris as a work sample for the Datarails Forward Deployed Financial Engineer, Applied AI application. Not affiliated with or endorsed by Datarails, not a Datarails product, and it does not describe how any Datarails product works.

Rerun Desk: the same Q3 variance question, asked again a week later

Which cost centers need commentary for Q3, and why?

Sample Co (synthetic) locked its Q3 variance commentary as Q3-v1 on close day 5. A week later the question is asked again. Two inputs have moved: journal J-1004, a $6,000 legal accrual to G&A posted on close day 9, and mapping M2, which rolls Data Platform into Engineering. Code computes every figure; a model drafts sentences at build time from a facts object; a numbers gate checks them; a person classifies, signs and locks. Policy: flag when the rounded percent is above 5.0% and the variance is at least $10,000.

1. Rerun: Q3-v1 (S1, M1, budget) to Q3-v2 (S2, M2, budget)

Cost center	Q3-v1 variance	Q3-v2 variance	Attributed to (snapshot, then mapping)
Engineering	$36,000 (1.5%), not flagged	$54,000 (2.3%), not flagged	mapping M1 to M2: actual +$18,000, FTE +2
Sales	$112,500 (7.5%), flagged	$112,500 (7.5%), flagged	no change
Marketing	-$54,000 (-6.0%), flagged	-$54,000 (-6.0%), flagged	no change
G&A	$29,400 (4.9%), not flagged	$35,400 (5.9%), flagged	snapshot S1 to S2: actual +$6,000
Customer Success	$9,000 (6.0%), not flagged	$9,000 (6.0%), not flagged	no change
People	-$3,000 (-1.0%), not flagged	-$3,000 (-1.0%), not flagged	no change
Data Platform	$18,000, unplanned spend	retired into Engineering	mapping M1 to M2: actual -$18,000, FTE -2
Company total	$147,900 (2.5%)	$153,900 (2.6%)	snapshot +$6,000; mapping nets to $0

- G&A moves from $29,400 (4.9%) to $35,400 (5.9%) and now needs commentary; all of the change is J-1004, which a day-5 snapshot cannot contain.
- Engineering moves from $36,000 (1.5%) to $54,000 (2.3%) because Data Platform's $18,000 is now inside it; still below the threshold. Data Platform is retired into Engineering, so its unplanned-spend flag clears.
- 4 rows are identical (Sales, Marketing, Customer Success, People). Drafts: 2 reused with the same text (Sales, Marketing), 1 new (G&A), 1 retired (Data Platform).
- Recreate the prior version: replaying the Q3-v1 inputs against today's data returns the locked figures and SHA-256 f480e1ce2e0efcfd..., equal to the lock record. With regression R5 (the snapshot day ignored) the replay picks up J-1004, G&A shows 5.9% and flagged in a version that never contained it, the hash differs, and E15 blocks release.

2. One definition changed: plan basis budget to August reforecast (S1, M1)

Cost center	Budget plan	Reforecast plan	Budget basis	Reforecast basis	Test
Engineering	$2,400,000	$2,310,000	$36,000 (1.5%), not flagged	$126,000 (5.5%), flagged	should change
Sales	$1,500,000	$1,560,000	$112,500 (7.5%), flagged	$52,500 (3.4%), not flagged	should change
Marketing	$900,000	$870,000	-$54,000 (-6.0%), flagged	-$24,000 (-2.8%), not flagged	should change
G&A	$600,000	$600,000 (budget, no reforecast filed)	$29,400 (4.9%), not flagged	$29,400 (4.9%), not flagged	protected
Customer Success	$150,000	$150,000 (budget, no reforecast filed)	$9,000 (6.0%), not flagged	$9,000 (6.0%), not flagged	protected
People	$300,000	$300,000 (budget, no reforecast filed)	-$3,000 (-1.0%), not flagged	-$3,000 (-1.0%), not flagged	protected
Data Platform	$0	$18,000	$18,000, unplanned spend	$0 (0.0%), not flagged	should change
Company total	$5,850,000	$5,808,000	$147,900 (2.5%)	$189,900 (3.3%)	plan bridge -$42,000

The computed should-affect set (cost centers with a filed reforecast that differs from budget) is Engineering, Sales, Marketing, Data Platform. Changed rows: 4, all inside the set. Protected rows unchanged: 3 of 3. Actuals unchanged: 7 of 7. A missing reforecast falls back to budget and is labeled; with regression R6 (a missing reforecast read as zero) G&A, Customer Success and People become unplanned spend for their full actuals, protected rows change, and E16 blocks release.

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3. How it is checked

An answer key written by hand from the rules (10 scenarios, 2 expected diffs), an independent Python implementation written from the spec, seven gate tests, a lock hash, and a lineage check that recomputes every displayed figure from its formula and source fields. The same checks run on the page and in Node before every publish. Results on this build:

- E01 PASS: Sales flagged unfavorable, $112,500, 7.5%
- E02 PASS: Marketing flagged favorable, -$54,000, -6.0% (favorable variances are not ignored)
- E03 PASS: G&A at 4.9% not flagged in Q3-v1 (threshold decoy)
- E04 PASS: Customer Success not flagged at F = $10,000; flagged at F = $5,000
- E05 PASS: Engineering at 1.5% not flagged
- E06 PASS: Sales FTE reported as 45 against a plan of 42 (end of period, never summed)
- E07 PASS: People deferred while September is open: no variance, no percent, no draft; total incomplete, partial $150,900 (2.7%)
- E08 PASS: Sales and Marketing held while J-0917 is pending; both outcomes shown; no drafts
- E09 PASS: J-0917 approved: Sales $72,500 (4.8%), Marketing -$14,000 (-1.6%), neither flagged; rejected: as posted
- E10 PASS: Data Platform flagged as unplanned spend, $18,000, no percent
- E11 PASS: Threshold 7.0 with the default floor: flagged set is Sales and Data Platform only
- E12 PASS: 574 of 574 engine figures and diff checks match the independent Python implementation (10 scenarios, 2 diffs)
- E13 PASS: 7 of 7 gate tests return the expected answer
- E14 PASS: 24 of 24 displayed drafts pass the gate against current figures (24 AI, 0 rule-based, 0 AI drafts replaced after failing the gate)
- E15 PASS: Rerun: Q3-v1 to Q3-v2 attributed by step (snapshot $6,000, mapping $0); 4 rows identical; drafts 2 reused, 1 new, 1 retired; Q3-v1 replays to its lock hash
- E16 PASS: Definition switch, plan basis budget to August reforecast: 4 of 4 changed rows inside the computed should-affect set of 4; protected rows unchanged 3 of 3; actuals unchanged 7 of 7; plan bridge -$42,000
- E17 PASS: 500 of 500 displayed figures recompute from their formula and source fields across 10 scenarios

6 of 6 simulated regressions blocked, each with its named check failing: R1 E06, R2 E02, R3 E07, R4 E08, R5 E15, R6 E16 (FTE summed, favorable variances ignored, open month as zero, pending journal ignored, rerun reads the latest snapshot, missing reforecast as zero).

4. Where the model and the reviewer sit

gpt-5.4-mini-2026-03-17 drafted 12 of 12 cost center states at build time, 12 API calls, 5,830 input and 802 output tokens, estimated $0.0080 at an assumed list price. Drafts are cached under a hash of the facts the model saw, so a rerun reuses text where the facts did not change; this makes released text repeatable, not the model deterministic. The model never computes, classifies, sets a flag or runs in the browser. A person classifies each flagged variance, signs and locks; locking is refused while any row is deferred or held, a draft fails the gate, or a check fails.

5. Limits

- Synthetic data from one author, who also wrote the answer key. The results show the logic, attribution, refusals and checks on a known case, not accuracy on a real ledger, model quality on real data, or time saved.
- Mocked: the governed data source (a JSON file), snapshots (a posted-day filter), permissions, lock storage (a file written once; page locks are session-only) and sign-off identity (a typed name).
- The gate cannot catch an invented cause stated without numbers; the reviewer covers that. Ten scenarios do not establish coverage.
- Attribution order (close state, snapshot, mapping, plan basis, policy, journal decisions) is a design choice; it matters only when two changes touch the same row.

6. Sources and notice

The rerun test and the two checks (formula and source fields for a figure; change one definition and ask again) are taken from two public posts by Didi Gurfinkel, Datarails CEO (May and October 2026). The duty to define how solutions are tested and evaluated is from the role posting. The plan-basis switch is the author's analog for the definition example; the pass criteria are the author's design.

Every company, cost center, figure, headcount, journal and name in this brief is invented. Interactive version: russh.work/rerun-desk